

Support Vector Machine (SVM)

Linear SVM fundamentals

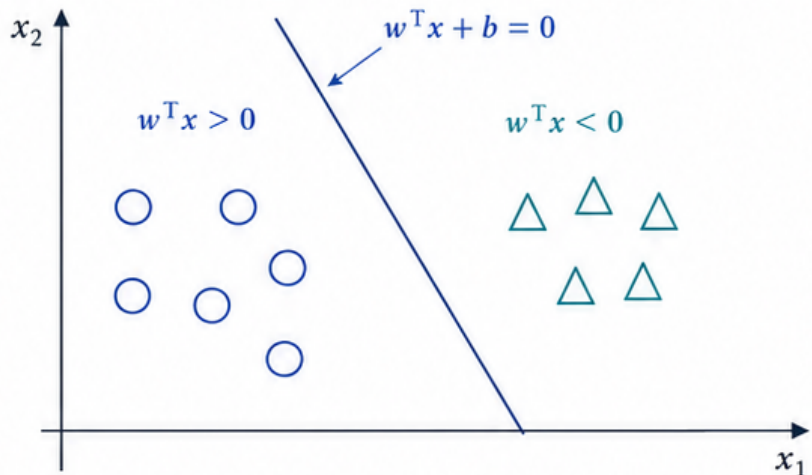


SVM is a supervised learning algorithm used for classification by finding the best separating hyperplane.

1 Linear SVM / Simple Linear Classifier

A linear classifier separates the feature space into two half-spaces using a hyperplane.

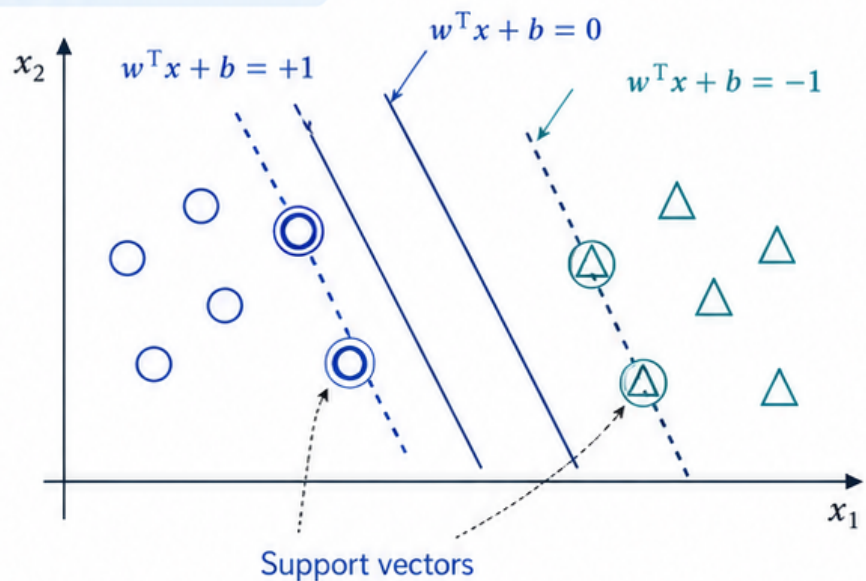
- Class +1
- △ Class -1



2 Concept of Support Vectors

The support vectors are the training points closest to the separating hyperplane. They lie on the margin boundaries and determine the position of the hyperplane.

- Class +1
- △ Class -1
- ⊙ Support vectors



3 Core equations

- Hyperplane: $w^T x + b = 0$
- If $w^T x + b > 0 \Rightarrow$ predict class +1
- If $w^T x + b < 0 \Rightarrow$ predict class -1

- $w = (w_1, w_2)$

- Margin width:

$$\frac{2}{\|w\|}$$

SVM Constraints and Example Dataset

Identifying the support vectors

1 Optimization constraint

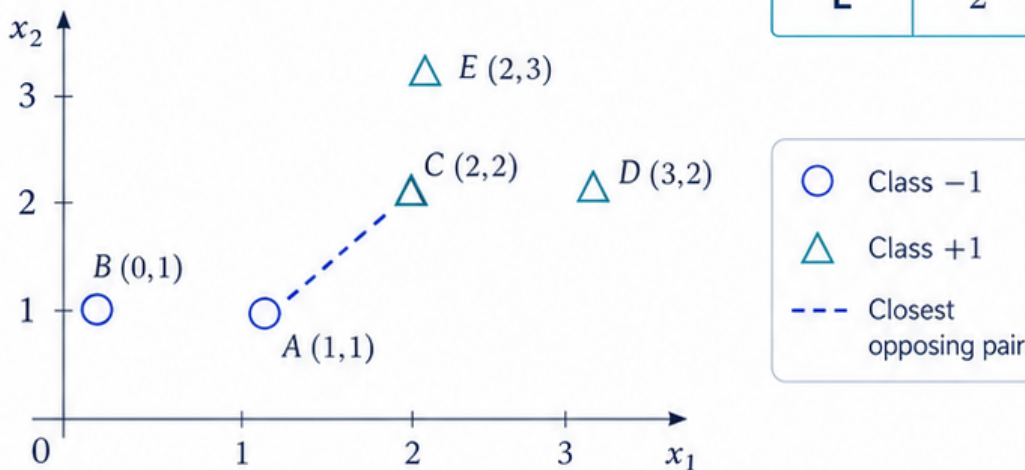
$$y_i (w^T x_i + b) \geq 1$$

This constraint ensures that all correctly classified points lie on or beyond the margin.

2 Example dataset

Point	x_1	x_2	y	Class
A	1	1	-1	○
B	0	1	-1	○
C	2	2	+1	△
D	3	2	+1	△
E	2	3	+1	△

3 Dataset visualization



4 Distance comparison

- $\text{dist}(A, C) = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{2} \approx 1.414$
- $\text{dist}(B, C) = \sqrt{(2-0)^2 + (2-1)^2} = \sqrt{5} \approx 2.236$
- $\text{dist}(A, D) = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5} \approx 2.236$

5 Conclusion



The closest opposing pair is A and C,
so A(1, 1) and C(2, 2) are the **support vectors**.

6 Direction

$$\text{Direction: } C - A = (2, 2) - (1, 1) = (1, 1)$$

Solving for w and b

Using the support vector conditions

- Since the direction between the support vectors is $(1, 1)$, let $w = k(1, 1)$.
- For each support vector, $y_i (w^T x_i + b) = 1$.

Key idea



The support vectors completely determine the separating hyperplane.

1 Support vector $C(2, 2)$, $y = +1$

$$\begin{aligned} (+1)(2k + 2k + b) &= 1 \\ 4k + b &= 1 \quad \dots \textcircled{1} \end{aligned}$$

2 Support vector $A(1, 1)$, $y = -1$

$$\begin{aligned} (-1)(k + k + b) &= 1 \\ 2k + b &= -1 \quad \dots \textcircled{2} \end{aligned}$$

3 Solve equations (1) and (2)

$$\begin{aligned} (4k + b) - (2k + b) &= 1 - (-1) \\ 2k &= 2 \\ k &= 1 \end{aligned}$$

4 Substitute back to find b

$$\begin{aligned} 4(1) + b &= 1 \\ b &= -3 \end{aligned}$$

5 Result



Therefore, $w = (1, 1)$ and $b = -3$.

Decision Boundary and Margin

Interpreting the linear SVM solution



$$w = (1, 1), \quad b = -3$$

1 Deriving the hyperplane

$$w^T x + b = 0$$

$$(1)x_1 + (1)x_2 - 3 = 0$$

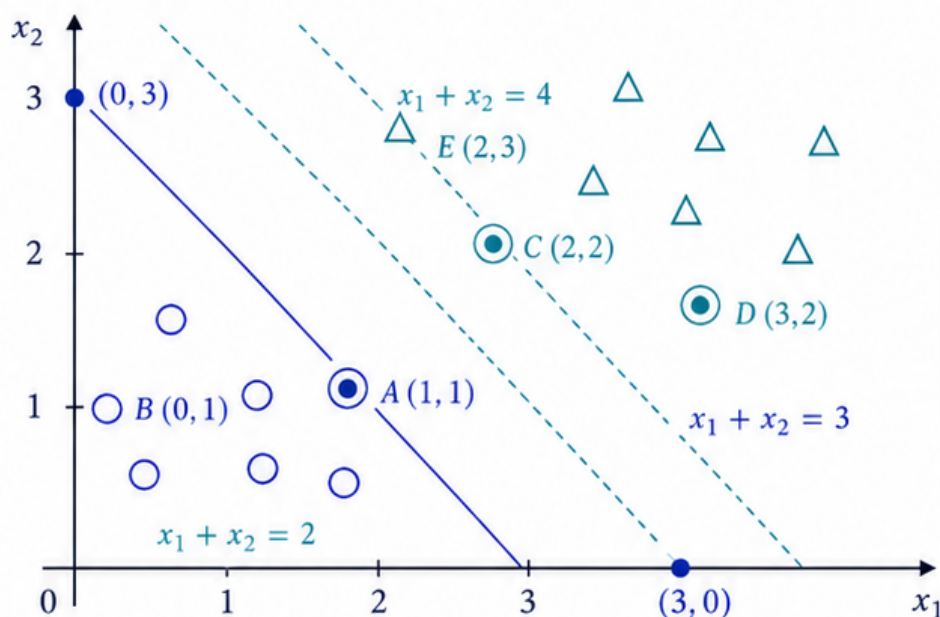
$$x_1 + x_2 - 3 = 0$$

This is the decision boundary.



Classification rule

- If $x_1 + x_2 > 3 \rightarrow$ class +1
- If $x_1 + x_2 < 3 \rightarrow$ class -1



- Class -1
- △ Class +1
- Support vectors

2 Distance from a point to the hyperplane

For any point $x = (x_1, x_2)$, the perpendicular distance to the hyperplane $x_1 + x_2 - 3 = 0$ is

$$\text{distance} = \frac{|x_1 + x_2 - 3|}{\sqrt{1^2 + 1^2}} = \frac{|x_1 + x_2 - 3|}{\sqrt{2}}$$

- For $A(1, 1)$:

$$\text{distance} = \frac{|1 + 1 - 3|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

- For $C(2, 2)$:

$$\text{distance} = \frac{|2 + 2 - 3|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$



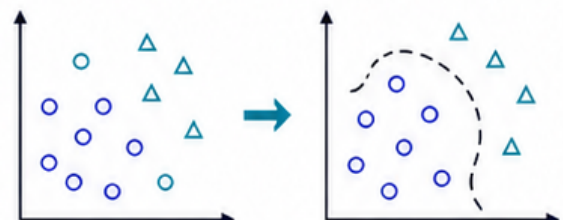
Both support vectors lie at equal distance from the decision boundary.

3 What if the data is not linearly separable?

When classes overlap, a single straight line cannot separate them.



SVM introduces slack variables and/or uses a kernel function to find a non-linear boundary.



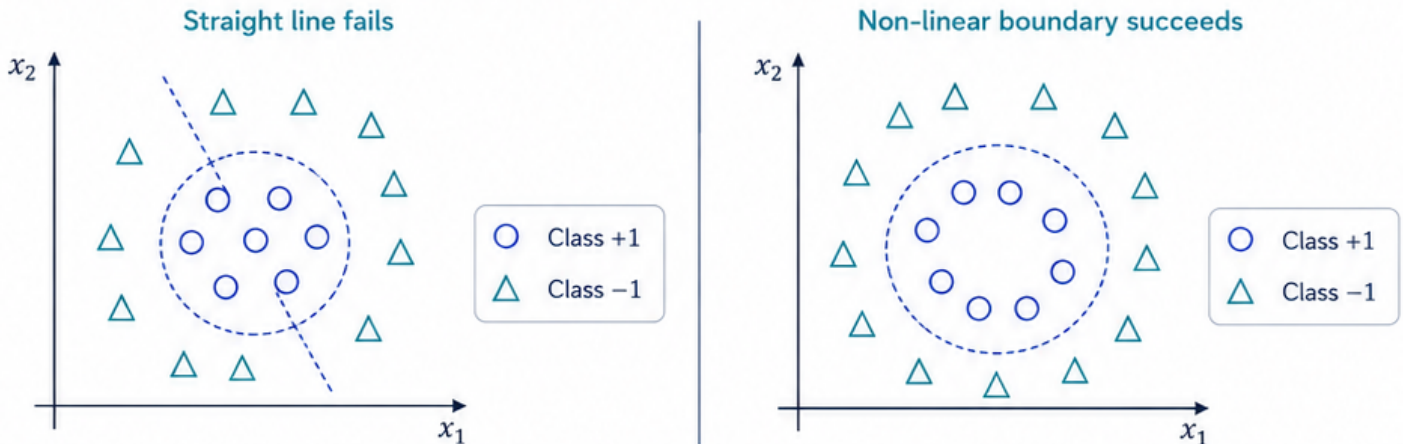
Non-linear SVM and Kernels

Handling data that is not linearly separable



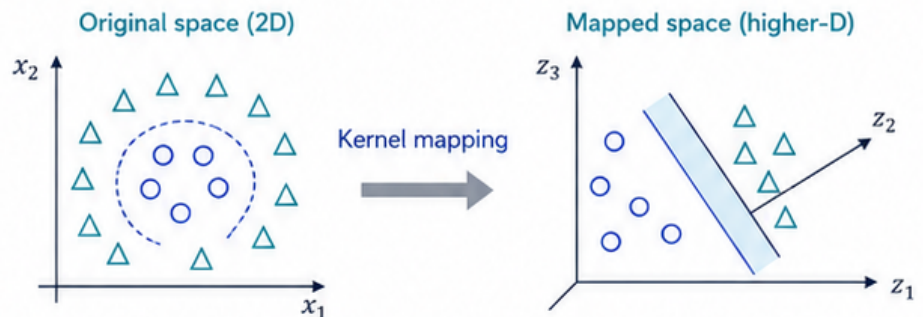
When a straight line cannot separate the classes, SVM can use kernel functions to create non-linear decision boundaries.

1 Non-linear SVM intuition



2 Kernel idea

A kernel implicitly maps the data to a higher-dimensional space where a linear separator becomes possible.



3 Common kernels

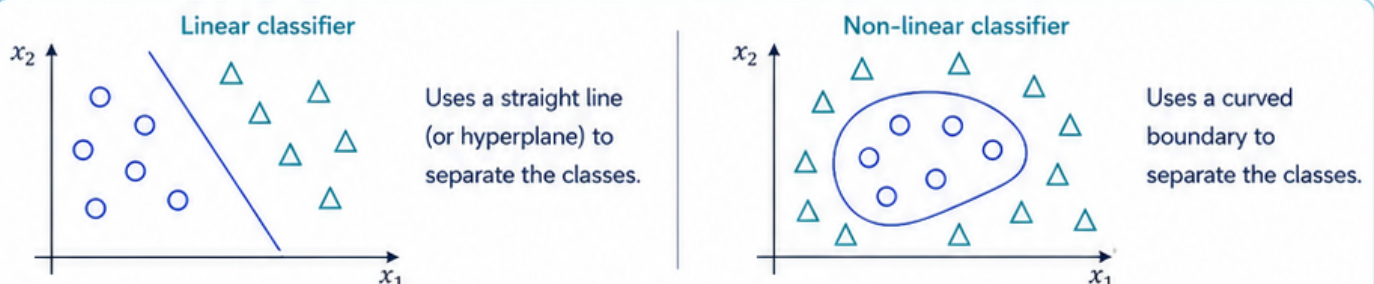
- **Linear kernel:** $K(\mathbf{x}, \mathbf{z}) = \mathbf{x}^T \mathbf{z}$
- **Polynomial kernel:** $K(\mathbf{x}, \mathbf{z}) = (\gamma \mathbf{x}^T \mathbf{z} + r)^d$
- **Radial Basis Function (RBF) kernel:**
 $K(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|^2)$

4 Key takeaway



Linear SVM works well when classes are separated by a line or hyperplane. Kernel SVM is useful when the boundary is curved.

5 Linear vs. Non-linear classifiers

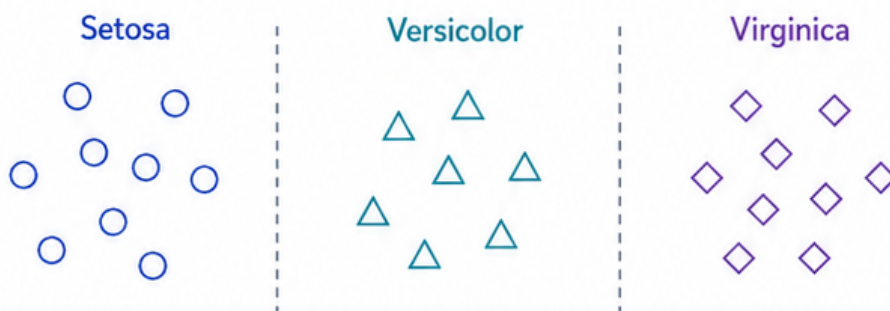


Multiclass SVM and Quick Recap

From binary classification to multiple classes

1 Multiclass strategies

- **One-vs-Rest (OvR):** train one classifier per class against all others.
- **One-vs-One (OvO):** train a classifier for each pair of classes.



One-vs-One (OvO) would use the pairs:

- Setosa vs Versicolor
- Setosa vs Virginica
- Versicolor vs Virginica

2 SVM summary

- Finds the maximum-margin separating hyperplane.
- Support vectors are the closest training points.
- For the example in these notes, the decision boundary is $x_1 + x_2 - 3 = 0$.
- Linear SVM handles linearly separable data.
- Kernel SVM handles non-linear data.

3 Key formulas

- Hyperplane:

$$w^T x + b = 0$$

- Constraint:

$$y_i(w^T x_i + b) \geq 1$$

- Margin width:

$$\frac{2}{\|w\|}$$



SVM is powerful because only a few support vectors determine the final classifier.